



Bootcamp on Intelligent Cyber Forensics: AI-Powered Investigation & Incident Response

Autonomous Network Intelligence & Security Assessment System

Cyber knights

KUNAL MADHAVAI

GAURAV TUWAR

SAMRUDDH SHELKE

JANHVAI PAWAR

DHANASHRI MADHAVAI



PROBLEM STATEMENT :

Create an AI-powered Autonomous Network Intelligence & Security Assessment System (ANISAS) that uses a single public IP address as input to automatically identify, evaluate, and display an authorized network's security posture via an intelligent risk assessment and interactive dashboard.

IMPORTANCE :

- Increasing cyber threats
- Faster risk identification
- Improved security decision-making



DOMAIN & RELEVANCE :

Domain

- Cybersecurity
- Network Security
- Threat Intelligence

Relevance

- Helps organizations identify security risks.
- Provides better visibility into network exposure.

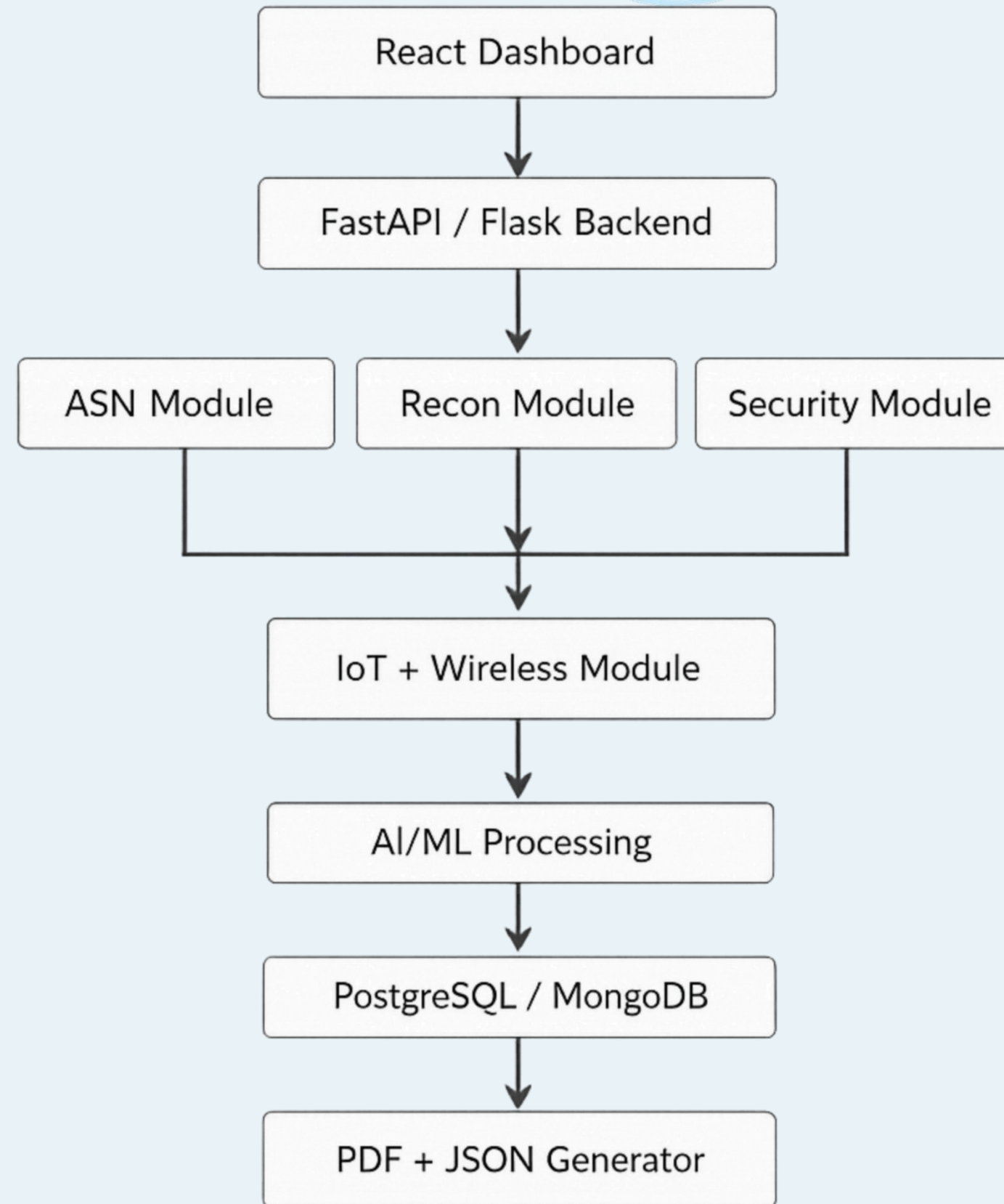


PROPOSED SOLUTION OVERVIEW :

- Accepts a public IP as input.
- Collects data from Nmap, Shodan, AbuseIPDB and NVD
- Performs AI-based risk analysis
- Generates interactive dashboard and reports.
- Provides security recommendations.



TECHNICAL WORKFLOW :





AGENTIC AI USED:

Role of AI

- Analyze scan results
- Correlate threat intelligence
- Prioritize vulnerabilities
- Generate executive summary
- Recommend security improvements



IMPLEMENTATION & TOOLS :

Frontend

- Next.js, React, TypeScript
- Tailwind CSS, Recharts

Backend

- FastAPI, PostgreSQL
- SQLAlchemy, WebSockets

Security Tools

- Nmap
- Shodan
- AbuseIPDB
- NVD API



OUTCOMES :

Achievements :

- Automated attack surface analysis
- Security & Risk Score calculation
- Device fingerprinting
- Interactive dashboard
- AI-generated security report
- PDF & JSON export



LIMITATIONS & FUTURE SCOPE :

Limitations

- Supports only authorized public IPs
- Depends on external APIs
- Active scanning requires permission

Future Scope

- Continuous monitoring
- Multi-IP scanning
- SIEM integration



CONCLUSION :

ANISAS provides an AI-powered and automated approach to network security assessment by integrating multiple intelligence sources into a single platform. It enables faster vulnerability identification, interactive visualization of security insights, and intelligent risk analysis through an intuitive dashboard. Its modular architecture also allows future enhancements, making it a scalable solution for improving an organization's overall security posture.